Md Kamal Uddin

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RESEARCH PROFILE

Experimental condensed-matter researcher specializing in the synthesis, doping, structural and optical characterization, and nonlinear optical properties of graphene-oxide-based and semiconductor nanocomposites, with applications in optical limiting and photocatalysis. Research experience includes rare-earth doping, Z-scan measurements, thermal lens spectroscopy, photoluminescence, UV–Vis spectroscopy, XRD/Rietveld refinement, and defect and band-gap analysis.

RESEARCH INTERESTS

- **Nonlinear optics of doped and defect-engineered materials**
- **Symmetry breaking and second-harmonic generation (SHG) spectroscopy**
- **Ultrafast light–matter interactions**
- **Rare-earth doping and defect states in functional nanomaterials**
- **2D materials and their optical properties**

EDUCATION

M.S. in Physics | University of Dhaka | 2019 – 2022
CGPA: 3.82/4.00
Thesis: “Synthesis of Magnetite (Fe_3O_4) Nanoparticles by Novel Thermal Decomposition Technique and Study Their Effect on Verwey Transition and Magnetic Fluid Hyperthermia (MFH).” | Supervisor: Prof. Dr. Ishtiaque M. Syed

B.S. in Physics | University of Dhaka | 2015 – 2019
CGPA: 3.85/4.00; ranked 2nd among 140 students

RESEARCH EXPERIENCE

Nonlinear Optics Research | Centre for Advanced Research in Science (CARS), University of Dhaka | Dec 2022 – Present

- Investigated third-order nonlinear optical properties of SnO_2–rGO, Nd-doped ZnO–rGO, and V_2O_5–GO nanocomposites using open- and closed-aperture Z-scan.
- Analyzed nonlinear responses using Sheik-Bahae and thermal lensing models; studied saturable and reverse saturable absorption, self-defocusing, and optical limiting.
- Migrated the Z-scan setup from LabVIEW to Python for automated instrument control and data acquisition.
- Upgraded the setup for simultaneous open- and closed-aperture data collection.
- Correlated nonlinear optical behavior with defect states using XPS, Raman, photoluminescence, and DFT calculations.
- Corresponding author on SnO_2–rGO study (*Materials Advances*, 2025; featured on journal cover).

Photocatalytic Nanocomposites Research | Department of Physics, University of Dhaka | 2024 – Present

- Contributed to synthesis and characterization of ZnO–rGO, GO–SnO_2, and Fe_3O_4–GO nanocomposites for wastewater treatment and antibiotic removal.
- Supervised two undergraduate research projects on composition- and defect-mediated photocatalytic enhancement.

M.S. Thesis Research | Department of Physics, University of Dhaka | 2021 – 2022
Supervisor: Ishtiaque M. Syed, Professor, Department of Physics, University of Dhaka
• Synthesized quasi-monodisperse superparamagnetic Fe_3O_4 nanocubes by thermal decomposition.
• Refined XRD data using FullProf and estimated particle size by Langevin function fitting.
• Identified Verwey transition signatures from ZFC/FC magnetization measurements.
• Evaluated magnetic hyperthermia performance via Néel and Brownian relaxation, achieving high specific loss power (published in *AIP Advances*, 2023).

Research Trainee | Materials Science Division, Bangladesh Atomic Energy Commission | 2019 – 2021
• Synthesized Fe_3O_4 nanoparticles by co-precipitation and characterized them using XRD, FTIR, TEM, DLS/zeta potential, and PPMS.
• Initiated a study of the Verwey transition, later continued in my M.S. thesis.

PUBLICATIONS (* corresponding author)

1. U. S. A. Meem, R. A. Jahan, **M. K. Uddin**, and I. M. Syed, "Functional development and enhanced photocatalytic efficiency of GO-supported SnO_2 by optimizing composition," *J. Mater. Sci.: Mater. Eng.* **21**, 34 (2026). https://doi.org/10.1186/s40712-026-00403-2
2. P. Das Gupta, **M. K. Uddin***, S. Barua, U. S. A. Meem, M. J. A. Sourov, T. Ahmed, M. Wahadoszamen, R. A. Jahan, and I. M. Syed, "Tuning the nonlinear optical properties of SnO_2–rGO nanocomposites: Exploration using conventional Z-scan and thermal lensing models," *Mater. Adv.* **6**, 9407–9426 (2025). https://doi.org/10.1039/d5ma00790a — *Featured on journal cover*
3. S. Jahan, M. A. Sayem, **M. K. Uddin**, F. Ferdous, R. A. Jahan, and I. M. Syed, "Remarkable photocatalytic performance and recyclable pathway of cleansing wastewater using ZnO–rGO nanocomposite," *AIP Adv.* **15**, 075343 (2025). https://doi.org/10.1063/5.0275565
4. **M. K. Uddin***, N. Deb, R. Rashid, H. Das, I. M. Syed, and S. M. Hoque, "Physical properties with high specific loss power of magnetite (Fe_3O_4) synthesized via thermal decomposition technique," *AIP Adv.* **13**, 105105 (2023). https://doi.org/10.1063/5.0164802

ONGOING RESEARCH & MANUSCRIPTS

- **Defect-Activated Strong Reverse Saturable Absorption in Nd^{3+}-Doped ZnO–rGO: Role of Defect States and Sequential Two-Photon Processes *(manuscript in preparation)***
- **Antibiotic removal from wastewater using Fe_3O_4-GO Nanocomposites. *(Submitted to Journal of Applied Physics)***
- **Nonlinear Properties of V_2O_5-GO using Z-scan technique. *(Data analysis in progress)***

PROFESSIONAL EXPERIENCE

Lecturer | Department of Physics, University of Dhaka | Dec 2022 – Present

- Teach undergraduate physics courses and conduct laboratory sessions. (Computational Physics, Electronics, Theory of Relativity, Undergraduate lab work)
- Supervise undergraduate research projects on photocatalytic nanocomposites.
- Develop course materials, examinations, and laboratory activities.

Lecturer | Department of Physics, BUET | Nov 2022 – Dec 2022

Taught undergraduate physics courses

Graduate Teaching Assistant | BRAC University | Oct 2021 – May 2022

Assisted with undergraduate physics courses and grading

TECHNICAL SKILLS

Nonlinear Optics: Z-scan (open/closed aperture), thermal lens spectroscopy, optical limiting

Materials Characterization: XRD, Raman, FTIR, UV–Vis, photoluminescence, XPS, FESEM, TEM, PPMS, DLS/zeta potential

Synthesis: hydrothermal, co-precipitation, thermal decomposition, rare-earth doping

Data Analysis & Refinement: FullProf, GSAS-II (Rietveld), OriginPro, MATLAB

Computational: VESTA, VASP (DFT)

Programming & Instrumentation: Python (instrument automation, data acquisition), C++, LaTeX

AWARDS AND HONORS

Dean's Honor Award (2024) | National Science and Technology Fellow (2020) | Semiconductor Technology Research Centre Fellow (2021) | National Talent Pool Scholarship (2019)

CONFERENCES & ACADEMIC ACTIVITIES

- **Oral presentation: "**Characterization and study of the magnetic hyperthermia effect of iron oxide (Fe_3O_4) nanoparticles synthesized via the thermal decomposition method**"** at the International Conference on Physics 2022, Dhaka
- Participant, Asian Science Camp 2017, Malaysia
- Participant, India-Bangladesh Winter School-2019 in High-Energy Physics

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TEST SCORES

- Physics GRE: 960 out of 990, 92nd percentile (2019) | IELTS Academic: 7.0 overall

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REFERENCES

- **Professor Dr. Ishtiaque M. Syed (Research Supervisor),** Department of Physics, University of Dhaka
  Email: imsyed@du.ac.bd | Phone: +880 1860808171
- **Dr. Engr. Sheikh Manjura Hoque (BAEC Research Supervisor),** Chief Scientific Officer and Head, Materials Science Division, Atomic Energy Centre, Dhaka
  Email: skmanjurahoque@gmail.com | Phone: +880 1712044155
- **Professor Dr. Md Wahadoszamen (Academic Referee),** Department of Physics, University of Dhaka
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